



The synergistic effect of poly(ethylene glycol)-borate ester on the electrochemical performance of all solid state Si doped-poly(ethylene glycol) hybrid polymer electrolyte for lithium ion battery

Dandan Li^a, Xiaoxiao Ji^a, Xianjing Gong^b, Fangchang Tsai^a, Qunchao Zhang^a, Li Yao^{a,**}, Tao Jiang^a, Robert K.Y. Li^b, Hengchong Shi^{c,***}, Shifang Luan^c, Dean Shi^{a,*}

^a Hubei Collaborative Innovation Center for Advanced Organic Chemical Materials, Ministry-of-Education Key Laboratory for the Green Preparation and Application of Functional Materials, Hubei Key Laboratory of Polymer Materials, School of Materials Science and Engineering, Hubei University, Wuhan, 430062, PR China

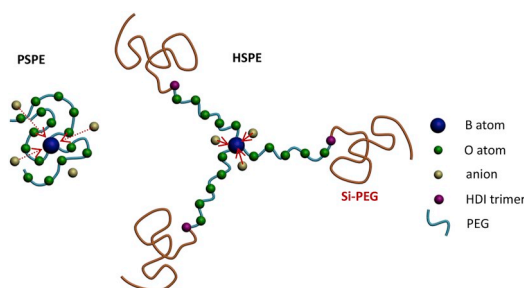
^b Department of Materials Science and Engineering, City University of Hong Kong, Tat Chee Avenue, Kowloon, Hong Kong, PR China

^c State Key Laboratory of Polymer Physics and Chemistry, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, Changchun 130022, PR China

HIGHLIGHTS

- HSPE: hybrid polymer electrolyte with B atoms covalently bonded into Si-PEG network.
- PSPE: B-PEG is added into Si-PEG network as a plasticizer.
- The B atoms in HSPE are more effective in trapping anions than that in PSPE.
- HSPE has much better electrochemical performance than that of PSPE.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

All solid-state polymer electrolyte
B-PEG
Si-PEG
Lithium ion transference number

ABSTRACT

Si doped-poly(ethylene glycol) (Si-PEG) hybrid polymer electrolyte with Si–O–C bonds in the main chain obtains high ionic conductivity even at room temperature because of the more flexible molecule chains. In this work, poly(ethylene glycol)-borate ester (B-PEG) is further covalently bonded into the crosslinked Si-PEG network to form the hybrid solid polymer electrolyte (HSPE). Comparing to previous works in which B-PEG is just added as a plasticizer, the Lewis acidity of boron (B) in the network of polymer chains is less shielded by the connected PEG chains, which is more effective to improve lithium ion transference number (t_{Li}^+) and reduce the interfacial polarization of LiFePO₄/HSPE/Li cells by attracting the anions of lithium salt. The t_{Li}^+ increases monotonously along with the increase of B-PEG contents in HSPE, while the ion conductivity has a peak value. When B-PEG content is 23.1 wt%, the highest ionic conductivity of the hybrid polymer electrolyte reaches up to 1.6×10^{-4} S/cm and t_{Li}^+ is 0.68 at 25 °C. And the initial capacity of LiFePO₄/HSPE/Li cells are 120 mAh/g and 90 mAh/g at 0.5C and 1C at 25 °C. After 100 cycles under 1C at 25 °C, the capacity retention of the LiFePO₄/HSPE/Li cell is 84%, which is higher than that of 65% for LiFePO₄/PSPE/Li cell. (PSPE indicates the Si-PEG polymer electrolytes with B-PEG just as plasticizer).

* Corresponding author.

** Corresponding author.

*** Corresponding author.

E-mail addresses: yaoli_2015@163.com (L. Yao), shihe@ciac.ac.cn (H. Shi), deanshi2012@hubu.edu.cn (D. Shi).

1. Introduction

Replacing the flammable liquid electrolyte [1–5] with solid ones [6–8] has been proved to be an effective way to increase the safety of liquid lithium ion batteries [9–13]. Compared with inorganic solid electrolyte, solid polymer electrolytes (SPEs) have many attractive properties, such as light weight, easy processing and excellent ductility, etc. [14–17]. However, high interfacial impedance and poor C-rate performance of lithium ion batteries with SPE largely inhibit its widely application. Accumulation and decomposition of the anions on the cathode surface during the cycling process result in an increase of interfacial impedance and a reduction of the cycle life of the lithium metal battery (LMB) [18]. Previous studies suggest that an increase of Li^+ transference number is crucial for the performance enhancement of LMB with SPE [19–24]. Lewis acids that can trap the anions, such as B and Al, are believed to be effective in increasing the Li^+ transference number. Molecular dynamics simulation results indicate that Li^+ diffusivity in polyborane is almost ten-fold higher than that in PEO and the anion diffusivities can be significantly decreased simultaneously [25]. Poly (ethylene glycol) (PEG)-borate ester (B-PEG) has been employed as a plasticizer in PEO or PAN based polymer electrolytes to enhance both the Li^+ transference number and the C-rate performance of the according LMBs in many previous studies [26–41]. However, although the addition of B-PEG plasticizer can increase the ionic conductivity by increasing the segments mobility [28,39], its function as a Lewis acid seems to be weakened because of the relatively low Li^+ transference number (0.2–0.5) [18,28,40].

In our previous work, Si doped flexible self-supporting comb like PEG-based copolymer (Si-PEG) is synthesized via condensation polymerization by using n-propyltrimethoxysilane (PTMS), polyethylene glycol monomethyl ether (MPEG) and PEG. The according LMB with such Si-PEG SPE can be cycled even at 10 °C [42]. In this work, combining the advantages of B-PEG and Si-PEG, HSPE is synthesized by covalent bonding B-PEG into the crosslinked Si-PEG network with hexamethylene-1,6-diisocyanate homopolymer (HDI trimer) as cross-linker [43,44]. For comparison, polymer electrolytes with B-PEG being added into Si-PEG matrix just as plasticizer are also prepared. This plasticized polymer electrolyte is abbreviated as PSPE in the following text.

2. Experimental works

2.1. Materials

PEG ($M_n = 400$ g/mol) was purchased from Sinopharm Chemical Reagent Co. Ltd., and MPEG ($M_n = 1000$ g/mol) from Tokyo Chemical Industry Co. Ltd. Both were dried in vacuum at 120 °C for 24 h before use. Glacial acetic acid and Toluene were purchased from Sinopharm Chemical Reagent Co. Ltd. Hexamethylene-1, 6-diisocyanate homopolymer (HDI trimer) (27.2% in butyl acetate solvent) was received from Bayer Material Science. n-propyltrimethoxysilane (PTMS) and boric anhydride (B_2O_3) were obtained from J&K company. Lithium bis (trifluoromethane sulfonimide) (LiTFSI) was purchased from Sigma-Aldrich. N,N-Dimethyl Formamide (DMF, 99.8%) was purchased from Aladdin. Olivine type LiFePO_4 (The average size of LiFePO_4 was about 800 nm) and acetylene black were purchased from Shenzhen Kejing Star Technology Co., Ltd. Polyvinylidene fluoride (PVDF $M_w = 1,000,000$ g/mol) was received from Arkema. These materials were used as received. Tetrahydrofuran (THF) was refluxed with

Table 1

Polymer electrolytes used in this work.

Sample	Si-PEG(g)	B-PEG(g)	HDI trimer (mol)	B-PEG content (wt. %)
HSPE-1	1	0	0.002	0
HSPE-2	1	0.2	0.003	16.7
HSPE-3	1	0.3	0.0034	23.1
HSPE-4	1	0.5	0.004	33.3
HSPE-5	1	1	0.006	50
HSPE-6	1	2	0.008	66.7
HSPE-7	0	1	0.01	100
PSPE	1	0.3	0.005	23.1

calcium hydride (CaH_2) overnight, followed by reduced pressure distillation.

2.2. Synthesis of HSPE polymer

Si-PEG was synthesized according to our previous method [42]. PEG (12 g, 0.03 mol), MPEG (10 g, 0.01 mol) and PTMS (4.92 g, 0.03 mol) were stirred in a 250 mL three neck flask for 30 min at 80 °C. After that, glacial acetic acid was added to adjust the PH value to 3–5 and further stirred for 6 h at 80 °C. The side product methanol was removed by reduced pressure distillation. Finally, 25 ml of THF was added into the flask to terminate the reaction. The resulted Si-PEG solution was denoted as **A** solution.

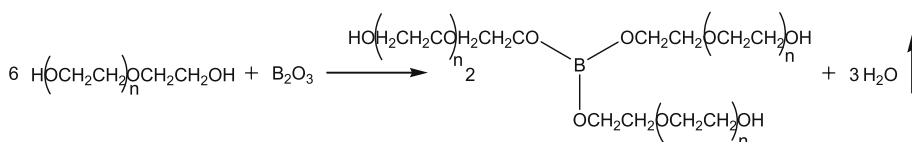
B-PEG was synthesized from the reaction of PEG and B_2O_3 , the process was shown in Scheme 1 [45]. PEG (26.4 g, 0.066 mol) and B_2O_3 (0.77 g, 0.011 mol) were mixed in a certain amount of toluene under N_2 atmosphere and refluxed at 110 °C to eliminate the generated water. After the completion of the reaction, toluene was evaporated under vacuum condition. Finally, the final product was dissolved in 10 ml THF. The solution was denoted as **B** solution.

Certain amount of THF solution containing HDI trimer was denoted as **C** solution. HSPE was obtained by mixing different amounts of **A**, **B** and **C** solutions simultaneously in the ice bath firstly. Then the temperature was increased to 60 °C to induce the reaction between hydroxyl groups on PEG and isocyanate groups on HDI. In this case, both Si-PEG and B-PEG should be crosslinked into the whole HSPE network. As a result, borane atoms were fixed in the polymer backbones. HSPE polymer electrolyte film with different weight ratio of Si-PEG and B-PEG were obtained by casting the HSPE solution in a PTFE plate to evaporate the solvent. The weight ratios of Si-PEG and B-PEG for HSPE-1, HSPE-2, HSPE-3, HSPE-4, HSPE-5, HSPE-6, HSPE-7 samples were 1:0, 1:0.2, 1:0.3, 1:0.5, 1:1, 1:2, 0:1, respectively.

For comparison, only **A** and **C** solutions were firstly mixed in the ice bath when PSPE was synthesized. **B** solution was added after the reaction between hydroxyl groups on Si-PEG and isocyanate groups on HDI was almost finished. In this case, the network with only Si-PEG was formed. B-PEG had no covalent bond connected to the Si-PEG network and existed freely in the SPE film. As the molecular weight of PEG used in this work is quite low ($M_n = 400$ g/mol), the molecular weight of B-PEG must be low. This is why its behavior in the SPE film is described as a plasticizer. All the specific prescriptions of the polymer electrolytes used in this paper are listed in Table 1.

2.3. Preparation of polymer electrolyte

Certain amount of the bis(trifluoromethane) sulfonimide lithium



Scheme 1. Synthesis scheme of B-PEG.

salt (LiTFSI) was dissolved in the obtained HSPE and PSPE solutions, respectively. The $[\text{Li}^+]:[\text{EO}]$ mole ratio was fixed at 1:20. The mixture was stirred for 12 h to form a homogenous colloid. The above mixture was heated at 80 °C for 6 h to evaporate the solvent and form a transparent membrane on a Teflon substrate. The thicknesses of HSPE and PSPE films in the cells are all around 50 μm .

2.4. Preparation of composite cathode

LiFePO_4 active material, acetylene black, polymer electrolyte and PVDF solution in DMF with unified weight ratio of 60:20:15:5 [46] were mixed for 48 h at 500 rpm. After that, the slurry was casted onto aluminum foil and dried at 100 °C for 72 h in the vacuum oven to remove DMF solvent. Finally, the composite cathode sheet was pressed under 10 MPa at room temperature.

2.5. Characterization

^1H nuclear magnetic resonance (^1H NMR) spectra were recorded using a UNITY INOVA 600 MHz Spectrometer (Varian Company Ltd.) at an ambient temperature with tetramethyl silane as the standard. ^{29}Si nuclear magnetic resonance (^{29}Si NMR) spectra were recorded using a UNITY INOVA 600 MHz Spectrometer (Varian Company Ltd.) at an ambient temperature with tetramethyl silane as the standard. Thermogravimetric analysis (TGA) data were achieved by Diamond TG/DTA from Perkin-Elmer. Samples weighing 5–10 mg were heated from 0 to 800 °C at a rate of 10 °C/min under nitrogen atmosphere.

Electrochemical impedance spectroscopy (EIS) measurements were made in the frequency range of 0.1–10⁶ Hz by applying an AC voltage with 10 mV perturbation via an electrochemical workstation (Shanghai Chenhua Instrument Co., Ltd) from 20 °C to 80 °C. The HSPE electrolyte membrane was sandwiched between two stainless steel block electrodes. The ionic conductivity of the electrolyte can be calculated from the equation: $\sigma = \frac{L}{S \cdot R}$, where L is membrane thickness, S represents effective electrode area, and R is the bulk resistance. Linear sweep voltammetry (LSV) measurement was carried out at the electrochemical workstation. The experiment was recorded from 0 V to 7 V at a scan rate 1 mVs⁻¹. The electrochemical stability window was evaluated using Li/HSPE/Li. The cell was assembled in a dry glove box in argon atmosphere.

The lithium ion transference number (t^+) can be defined as the ratio of the charge conducted by the lithium ion through a fixed reference plane to the total Faraday charge flowing through the plane, i.e.:

$$t^+ = \frac{i^+}{i^+ + i^-} \quad (1)$$

Where i^+ , i^- are the Faraday current portions conducted by the cations and the anions, respectively. The measurement of lithium ion

transference number t_{Li}^+ is carried out by using AC impedance spectroscopy and chronoamperometry (CA). A symmetric non-blocking cell, Li/Polymer Electrolyte/Li, is used to apply a small constant polarization voltage at the cell end to obtain a polarization curve of current versus time. If only concentration polarization process is taken into account, the lithium ion transference number (t^+) can be calculated as follows: At the beginning, there is no concentration polarization within the battery, and the initial current (I_0) is determined by the migration of anions and cations under an electric field. Since the anion in the polymer electrolyte does not participate in the electrode reaction at both ends of the electrode, the concentration gradient of the anion is gradually established in the battery with the extension of time. Therefore, the current gradually decreases to a steady value (I_s) when the net transportation of the anion is zero. In this case, the steady current (I_s) should be only contributed by the migration of lithium ions. Therefore, by definition, the transference number of lithium ion should be calculated by equation (2):

$$t^+ = \frac{I_{ss}}{I_0} \quad (2)$$

However, when the non-concentration polarization effects are taken into consideration, the initial current (I_0) and steady-state current (I_s) should be determined by equations (3) and (4), respectively:

$$I_0 = \frac{\Delta V}{R_0 + k/\sigma} \quad (3)$$

$$I_{ss} = \frac{\Delta V}{R_{ss} + k/t_{\text{Li}}\sigma} \quad (4)$$

Where: ΔV is the polarization voltage (V) applied to both ends of the battery, k is the cell constant (cm^{-1}), σ is the electrolyte conductivity (S/cm), and R_0 is the initial interface impedance of the battery (ohm), R_{ss} is the interface impedance after the battery is polarized (ohm). By combining equations (3) and (4), equation (5) should be finally used to calculate the lithium ion transference number:

$$t_{\text{Li}}^+ = \frac{I_{ss}(\Delta V - I_0 R_0)}{I_0(\Delta V - I_{ss} R_{ss})} \quad (5)$$

All the tests (ionic conductivity and lithium transference number) are carried out 10 times with different piece of samples and the average values are calculated and recorded as the results in the paper.

3. Results and discussion

Fig. 1a shows the ^{29}Si NMR spectra of PTMS and Si-PEG. The chemical shifts of ^{29}Si in PTMS monomer are -41.8 and -42.3 ppm and in Si-PEG are -59.3 and -60.7 ppm, which is in accordance to the literature reported results [42,47] that the peak of ^{29}Si in PTMS will move

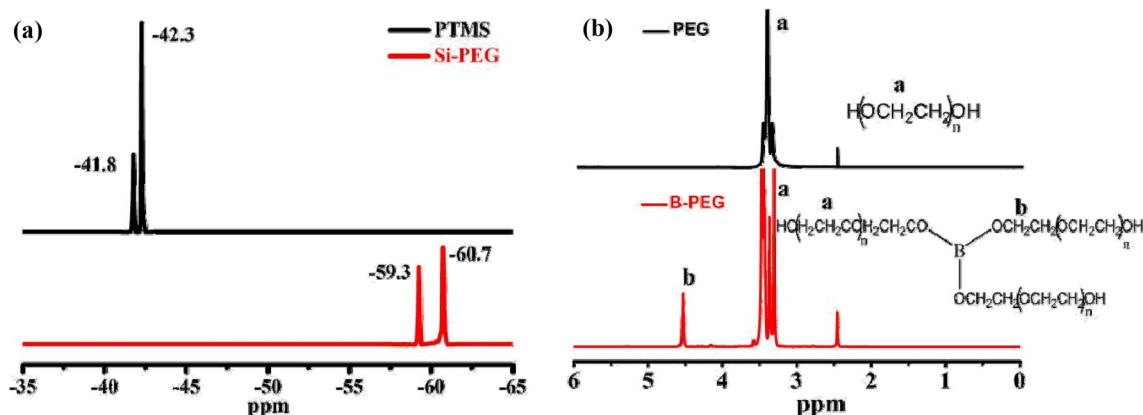


Fig. 1. (a) ^{29}Si NMR spectrum of PTMS and Si-PEG. (b) ^1H NMR spectrum of PEG and B-PEG.

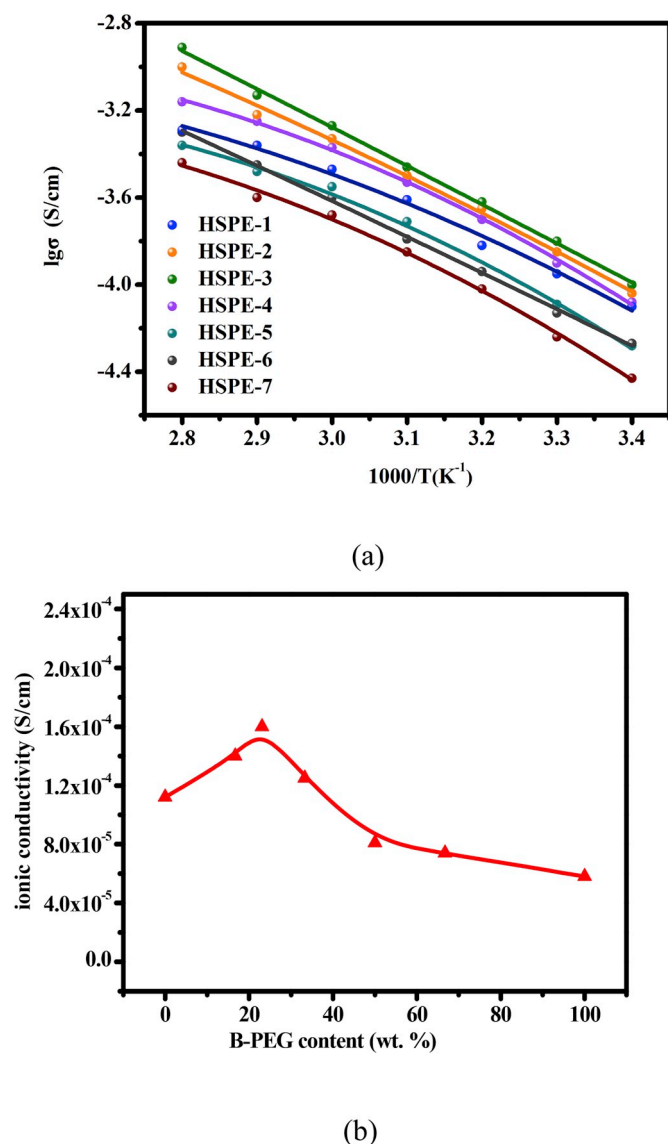


Fig. 2. (a) Temperature dependence of ionic conductivity of HSPE-1, HSPE-2, HSPE-3, HSPE-4, HSPE-5, HSPE-6 and HSPE-7. (b) The variation diagram of ionic conductivity of different SPEs with the raise of B-PEG at 25 °C.

to lower chemical shift after the polycondensation. Fig. 1b shows the 1H NMR spectra of PEG and B-PEG. The chemical shift at 2.5 ppm is the solvent of Dimethyl Sulphoxide (DMSO). The chemical shift at 3.4–3.5 ppm is attributed to the hydrogens of the $-CH_2-$ groups of PEG. The new peak at 4.5 ppm, which belongs to the $-CH_2-$ groups linked to the boronate center demonstrates the formation of B-PEG [48,49]. This result demonstrates that the B-PEG is synthesized successfully.

Fig. 2a illustrates temperature dependency of the ionic conductivities of HSPE-1, HSPE-2, HSPE-3, HSPE-4, HSPE-5, HSPE-6 and HSPE-7 at various temperatures ranging from 20 to 80 °C. Obviously, the higher the temperature is, the higher are the ionic conductivities for all these polymer electrolytes. In Fig. 2b, the ionic conductivities of HSPE first increase and then decrease with the increase of B-PEG content. At 25 °C, the highest ionic conductivity, 1.6×10^{-4} S/cm can be obtained when Si-PEG:B-PEG is 1:0.3 (HSPE-3), even higher than that of HSPE-1 (pure Si-PEG without B-PEG). The reason why the addition of certain amount B-PEG into the polymer can enhance the ionic conductivity might because of its ability to attract anions and promote the dissociation of lithium salt. However, when the content of B-PEG is too

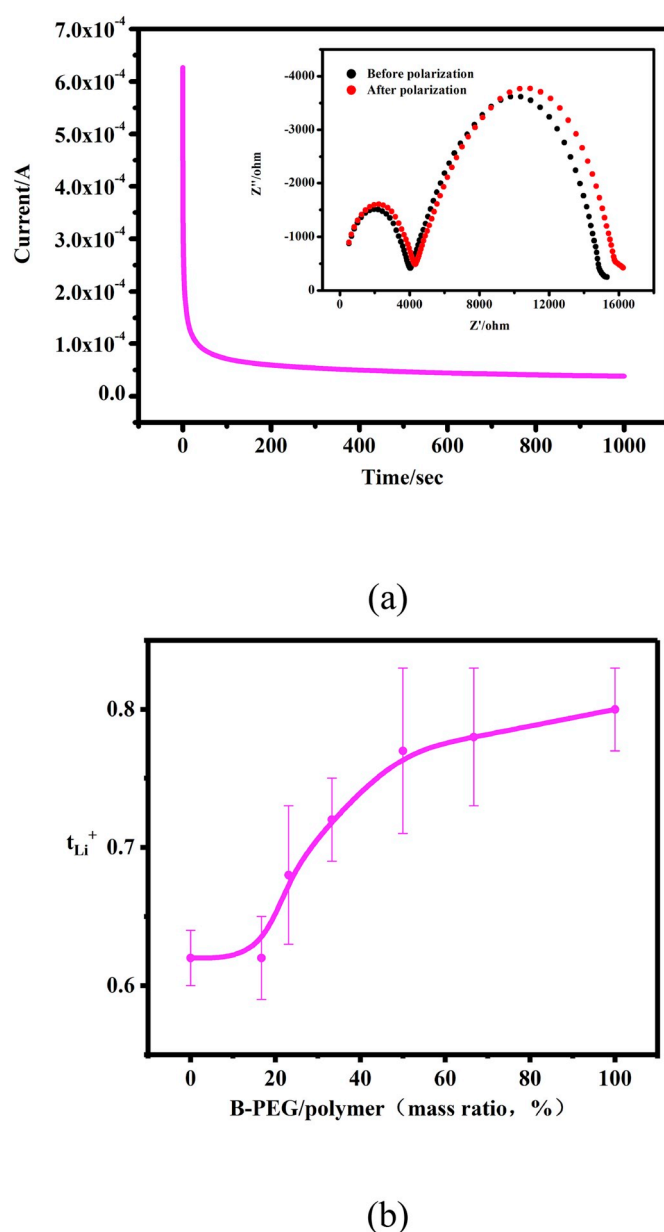


Fig. 3. (a) Variation of current with time during polarization of a Li/HSPE-3/Li, configuration at an applied voltage of 10 mV; the inset figure shows the impedance spectra before and after polarization at 25 °C. (b) The t_{Li}^+ variation of different SPEs with the raise of B-PEG contents at 25 °C.

high, the flexibility of polymer chains as well as the ionic conductivity of the polymer electrolyte will decrease. At an extreme, the ionic conductivity of B-PEG based copolymer electrolyte without Si-PEG (HSPE-7) is only 5.8×10^{-5} S/cm at 25 °C.

The variation of electric current with time and the impedance spectra before and after polarization for HSPE-3 is shown in Fig. 3a (Similar results for other polymer electrolytes are also got and not shown here). The t_{Li}^+ value is calculated according to equation (5) [50]. In Fig. 3b, the t_{Li}^+ values vary from 0.62 for HSPE-1 (without B-PEG) to 0.8 along with the increase of B-PEG contents. After the contents of B-PEG exceed 50 wt%, the t_{Li}^+ value does not change much from HSPE-5 to HSPE-7.

LiFePO₄/HSPE/Li cells with HSPE and PSPE are assembled and the according rate performances at 25 °C are examined. The initial discharge specific capacities of LiFePO₄/HSPE/Li and LiFePO₄/PSPE/Li cells with the rate of 0.5C and 1C at 25 °C are shown in Fig. 4. It can be

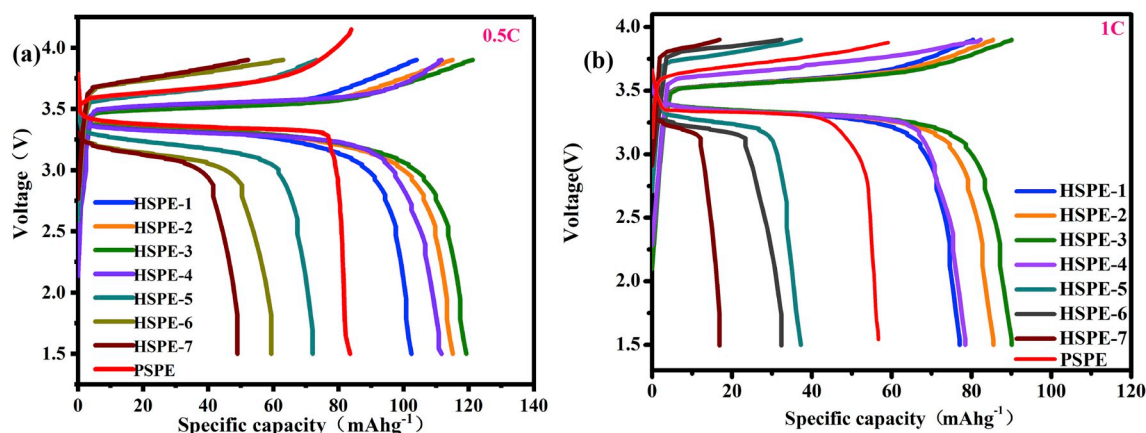


Fig. 4. (a) The initial charging-discharging specific capacity of LiFePO₄/HSPE/Li cells with HSPE-1, HSPE-2, HSPE-3, HSPE-4, HSPE-5, HSPE-6 and HSPE-7 and LiFePO₄/PSPE/Li cells with 0.5C at 25 °C. (b) The initial charging-discharging specific capacity of LiFePO₄/HSPE/Li cells with HSPE-1, HSPE-2, HSPE-3, HSPE-4, HSPE-5, HSPE-6 and HSPE-7 and LiFePO₄/PSPE/Li cells with 1C at 25 °C.

found that the initial discharge capacities of LiFePO₄/HSPE/Li cells first increase then decrease along with the increasing the B-PEG content in the system, which follows the same varying trend of the ionic conductivity shown in Fig. 2. The initial discharging specific capacities are only 50 mAhg⁻¹ at 0.5C and 13 mAhg⁻¹ at 1C for LiFePO₄/HSPE-7/Li cell, where there is no Si-PEG in the polymer electrolyte. At the same time, HSPE-3 has higher initial discharging specific capacities than PSPE while both electrolytes have the same content of B-PEG. This result proves that the existence of Si—O—C flexible chains in the polymer electrolyte is the key factor to guarantee the high ionic conductivity of polymer electrolytes. Furthermore, synergistic effects of Si and B atoms in the main chain network are indispensable in HSPE to get good electrochemical performance of the LiFePO₄/HSPE/Li batteries. At the same time, HSPE has higher capacity than PSPE in the first charge and discharge cycle while both have the same content of B-PEG.

In order to study the effects of the existence state of borane atoms on the ion conductivities and t_{Li}^+ values of the polymer electrolyte, the ionic conductivities of HSPE-3 and PSPE (where B-PEG is just added as a plasticizer) at temperatures ranging from 20 to 80 °C are illustrated in Fig. 5a. For both polymer electrolyte, the mass ratio of Si-PEG and B-PEG are the same as 1:0.3. However, the ionic conductivity for HSPE-3 is much higher than that of PSPE within the whole temperature range. And the t_{Li}^+ value for HSPE-3 (0.68) is also higher than that for PSPE (0.4) (Fig. 5b). This indicates that the synergistic effect of B-PEG on the electrochemical performance of polymer electrolyte is more prominent by introducing B atoms onto the main chain of polymer electrolyte rather than only as the plasticizer.

It is found from the i - t curves shown in Figs. 3a and 5b that the polarization degree of PSPE after polarization is even lower than that of HSPE. This seems contradictory to the result of t_{Li}^+ values for these two polymer electrolytes. The reason for this phenomenon should be explained as follows: During polarization process, migration ability difference of cation and anion under electric field will induce concentration polarization, which is directly related to the t_{Li}^+ value of the polymer electrolyte. However, some non-concentration polarization effects, such as the initial surface difference of lithium electrode [51] and/or passivation layer between lithium metal and polymer electrolyte [52], will also affect the polarization degree during the polarization process. Therefore, the whole polarization degree cannot be directly used to evaluate the t_{Li}^+ values of polymer electrolytes. Although some researchers reported that polymer electrolytes with high polarization degrees have relatively lower t_{Li}^+ values [53,54], examples with both high t_{Li}^+ values and high polarization degrees are also reported [55–57]. Furthermore, polymer electrolytes with small polarization degrees may also have high [58–63] and low [64–66] t_{Li}^+ values. This is why Equation (5) other than Equation (2) is used to calculate the t_{Li}^+ values.

The cycling performance for LiFePO₄/SPE/Li cells with HSPE-3 and PSPE at 25 °C/0.5C and 25 °C/1C are shown in Fig. 6a and b. At 25 °C/0.5C, capacities retention of LiFePO₄/HSPE-3/Li and LiFePO₄/PSPE/Li cells are 91.3% and 81.5% after 100 cycles. While at 25 °C/1C, capacities retention of LiFePO₄/HSPE-3/Li and LiFePO₄/PSPE/Li cells are 84% and 65% after 100 cycles. It is clear that LiFePO₄/HSPE-3/Li cell has much better cycling properties than LiFePO₄/PSPE/Li cell. In Fig. 6c, when the LiFePO₄/HSPE-3/Li and LiFePO₄/PSPE/Li cells are

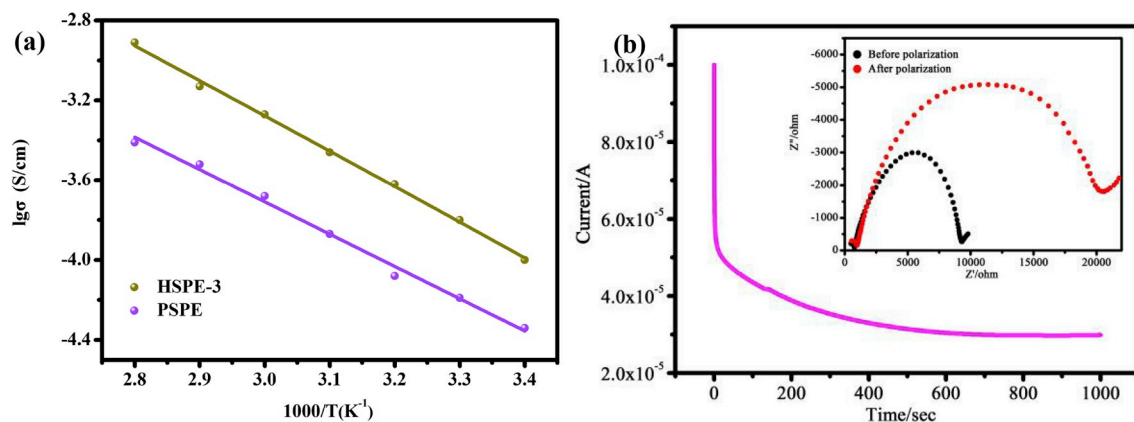


Fig. 5. (a) Temperature dependence of ionic conductivity of HSPE-3 and PSPE. (b) Variation of current with time during polarization of Li/PSPE/Li configuration at an applied voltage of 10 mV; the inset figure shows the impedance spectra before and after polarization at 25 °C.

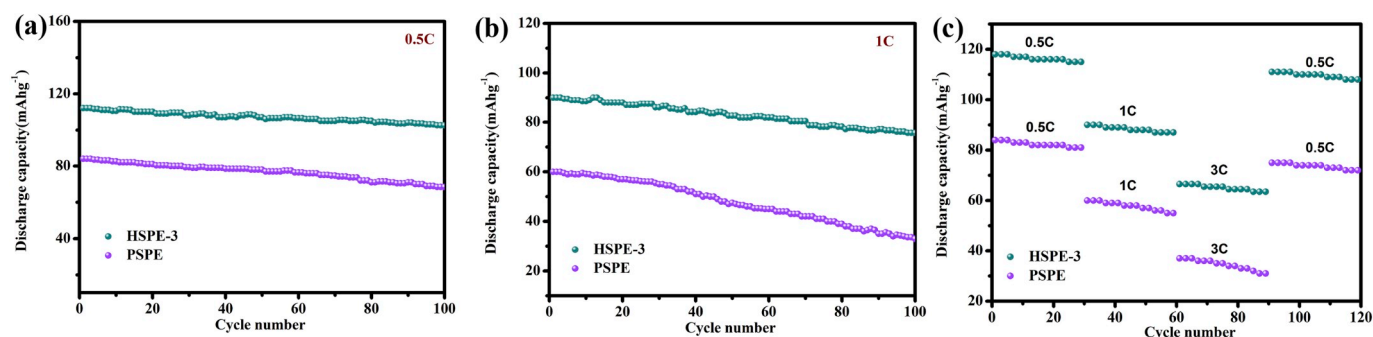
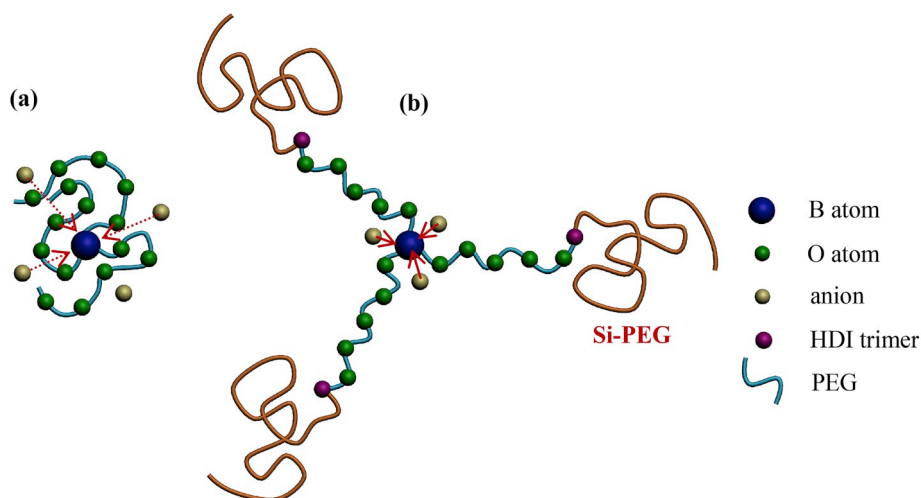
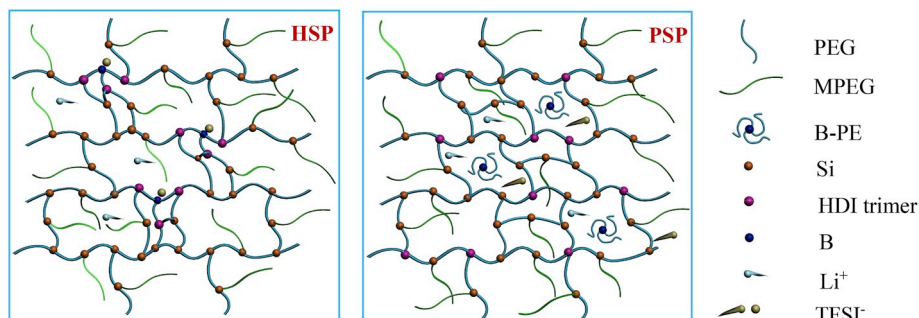


Fig. 6. The cycle performance of $\text{LiFePO}_4/\text{HSPE-3}/\text{Li}$ and $\text{LiFePO}_4/\text{PSPE}/\text{Li}$ cells at 0.5C (a) and 1C (b) current rate at 25 °C. (c) Rate capability of $\text{LiFePO}_4/\text{HSPE-6}/\text{Li}$ and $\text{LiFePO}_4/\text{PSPE}/\text{Li}$ cells at 25 °C.



Scheme 2. (a): Structure of B-PEG as plasticizer. (b): Structure of B-PEG when it is connected into the network of the network of polymer electrolyte.



Scheme 3. Structure of HSPE and PSPE polymer network and the migration of Li^+ , TFSI^- .

running from 0.5C to 3C then back to 0.5C at 25 °C, there are no obvious capacity deterioration in both cells. However, the according capacities of $\text{LiFePO}_4/\text{PSPE}/\text{Li}$ cell with B-PEG as plasticizer are much lower than those $\text{LiFePO}_4/\text{HSPE-3}/\text{Li}$ cells with B-PEG located in the main chain of polymer electrolytes. These results indicate that the synergistic effect of Si-PEG and B-PEG in enhancing the electrochemical performance is more prominent when both Si and B atoms are attached in the main chain of PEG network.

Why B-PEG has relatively low synergistic effect when it acts as a plasticizer? The possible reason might be that the borane atoms in B-PEG plasticizer are easily wrapped by the surrounding oxygen atoms in PEG chains, which makes them cannot interact with the anions of lithium salts effectively, as shown in Scheme 2a. If borane atom is connected into the network of the PEG networks, as shown in Scheme 2b, where the jointed PEG chains are more stretched and the borane atoms can be

exposed to trap the anions. In this case, the Li^+ transference number should be increased. Indeed, moderate trapping effects of B on anions (HSPE-1, HSPE-2 and HSPE-3) will also favor the dissociation of lithium salt which will result in increase of ionic conductivities as well as the whole electrochemical performance of the polymer electrolyte. However, if the trapping effect is too strong (HSPE-4, HSPE-5, HSPE-6 and HSPE-7), although high Li^+ transference numbers can be obtained, the ionic conductivities will decrease sharply. The according electrochemical performance of the SPE will decrease. The whole network structures of HSPE and PSPE are shown in Scheme 3. It is found that B-PEG is free out from the Si-PEG network in PSPE, while it is covalently bonded into the Si-PEG network in HSPE.

In Fig. 7a, the TGA curves show the thermal stability of HSPE-3 is higher than that of PSPE. And the initial decomposition temperature of HSPE-3 reaches up to 300 °C. This result confirms that HSPE-3 is

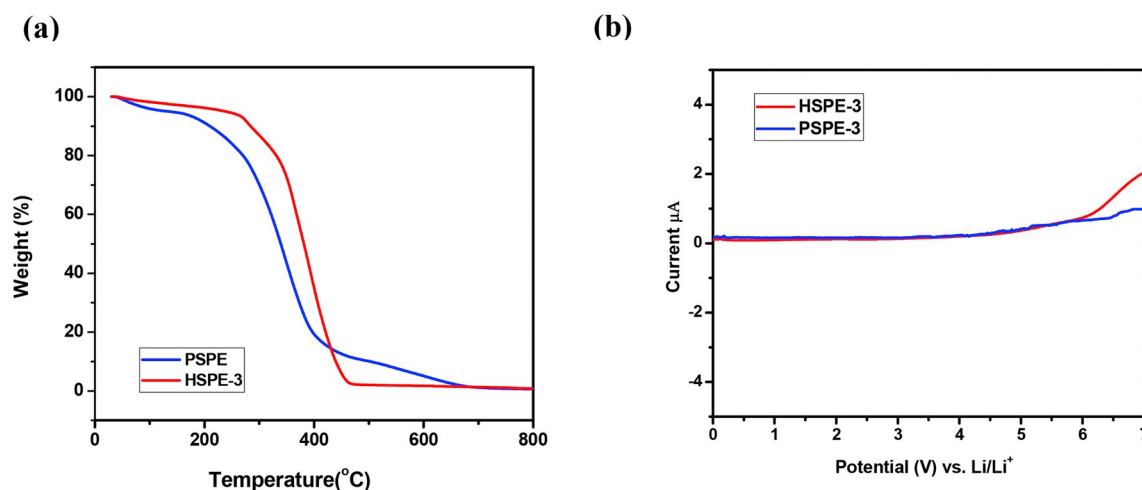


Fig. 7. (a) TGA curves of HSPE-3 and PSPE. (b) Linear sweep voltammograms of a Li/HSPE-3/Li and Li/PSPE/Li configuration at 25 °C.

Table 2

Comparison of the electrochemical performances of polymer electrolyte with different molecule structure.

Sample	Temperature/°C	Molecule structure	t_{Li}^+	Capacity*(mAh/g)/C-rate	Ref.
SPE-1	60	PDE600-PME4000-BPEG	0.34	–	40
SPE-2	60	PEGMA-BPEG	0.34	–	18
SPE-3	60	Composite electrolyte PEO-LATP-BPEG	0.49	94.2/2C	28
SPE-4	30	PEO-Al ₂ O ₃ -BPEG	–	96/1C	39
SPE-5	25	BPEG-PVDF	–	84/0.5C	41
SPE-6	25	POSS-g-PEGMEM/B-PEGMA	0.58	131.1/0.5C	67
HSPE-7	25	B-PEG without Si-PEG	0.80	13/1C	This work
PSPE	25	Si-PEG with B-PEG as plasticizer	0.4	60/1C	This work
HSPE-3	25	Si-PEG and B-PEG as main chain	0.68	90/1C	This work

Discharge specific capacity* from LiFePO₄/SPE/Li cells.

thermodynamic stable and safe within the working temperature range of LMBs as solid polymer electrolyte. Fig. 7b represents the result of a linear sweep voltammetry (LSV) scan of HSPE-3 and PSPE sandwiched between two Li metal working electrodes at 25 °C. There are no obvious oxidations taking place up to 4.5 V versus Li⁺/Li at the both polymer electrolytes, which demonstrates HSPE-3 can be used in high-voltage lithium batteries.

For comparison, some results obtained by the previous reported PEG based polymer electrolytes are listed in Table 2. It can be found that not only for our work (PSPE), t_{Li}^+ is smaller than 0.5 for all other polymer electrolytes with B-PEG as plasticizer (SPE-1, SPE-2, SPE-3 and SPE-4). While t_{Li}^+ of HSPE-7 is 0.8 and the t_{Li}^+ of HSPE-3 polymer electrolyte is 0.68 in our work when borane atoms are attached into the Si-PEG networks. Similar results obtained by Wei et al. [67] are deserved to be mentioned. In their work, borane atoms are attached onto the side chains of the POSS-g-PEGMEM/B-PEGMA polymer electrolytes. The t_{Li}^+ value can reach 0.58 and the battery has good cycling performance. Hence, exerting the synergistic effect of Si-PEG and B-PEG in polymer electrolyte is an effective strategy to improve the flexibility of molecule chains and get both high ionic conductivity and t_{Li}^+ of polymer electrolyte.

4. Conclusion

In summary, we prepare a flexible polymer electrolyte containing Si-PEG and B-PEG in the main chain and crosslinked via HDI trimer. The lithium ion transference number (t_{Li}^+) increases monotonically along with the increasing of B-PEG contents. However, the ion conductivities have a peak value. Taking both the values of ion conductivities and t_{Li}^+ into consideration, the optimal mass ratio of Si-PEG and B-PEG is 1:0.3. In this case, the ionic conductivity and t_{Li}^+ value of the polymer

electrolyte (HSPE-3) can reach 1.6×10^{-4} S/cm and 0.68 at 25 °C. LiFePO₄/HSPE-3/Li battery with HSPE-3 as polymer electrolyte has the best combination electrochemical performance. The initial capacity of LiFePO₄/HSPE-3/Li cell is 90 mAh/g under 1C at 25 °C and capacity retention is 84% after 100 cycles. This kind of polymer electrolyte shows productive electrochemical stability and thermodynamic stability. Synergistic effects of Si and B atoms in PEG based polymer electrolyte are more notable when both these two kinds of hybrid atoms are attached into the PEG networks. The existence of Si–O–C bonds will enhance the flexibility of molecule chains and get high ionic conductivity, while the Lewis acidic center, borane atoms will attract the anions of lithium salt to increase the value of t_{Li}^+ . When B-PEG is added just as plasticizer, the borane atoms are most likely to be shielded by the surround oxygen atoms in the attached PEG chains and cannot act as an anion trapper.

Acknowledgment

This work was supported by the NSFC No. 51173037, 51473047. The Doctoral Fund of Ministry of Education of China (No.20134208110006), and the Research Grants Council of the Hong Kong Special Administrative Region, China (Project Number: CityU 118313). Open Research Fund of State Key Laboratory of Polymer Physics and Chemistry, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences (201605), Hubei Key Project of Science and Technology (CXZD2017000293).

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